

Leveraging LLMs and Agentic AI for Enhanced Innovation and Organizational Decision Making

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Large language models (LLMs), agentic AI, innovation management, organizational decision-making, multi-criteria decision-making (MCDM), human–AI collaboration, explainable AI (XAI), decision support systems

Product innovation and early-stage strategic selection play crucial roles in enhancing organizational competitiveness. However, these processes face greater challenges due to fast-changing markets, technological uncertainty, and the increasing amount of diverse information. Managers and innovation teams must analyze complex data, consider multiple and sometimes conflicting criteria, integrate customer insights, and justify decisions in environments characterized by ambiguity and time pressure. Classical multi-criteria decision-making methods, i.e., AHP, TOPSIS, TRIZ, and MAUT, provide analytical hard times, but they struggle to accommodate the dynamic, unstructured, and iterative nature of innovation and early-phase decision processes. Similar limitations arise in innovation and opportunity exploration, where fragmented knowledge, incomplete customer understanding, and inconsistent reasoning often reduce transparency and hinder high-quality outcomes.

Recent advances in large language models (LLMs) and agentic AI architecture offer promising opportunities to augment these activities. LLMs can synthesize unstructured information, model customer perspectives, generate scenario-based insights, and support idea exploration. When organized into agentic configurations, specialized agents can collaborate on subtasks such as criteria formulation, scenario simulation, knowledge retrieval, customer preference modeling, and trade-off evaluation. These capabilities indicate potential to strengthen analytical depth, creativity, and transparency across innovation processes and organizational decisions. However, existing LLM-based and agentic approaches remain limited in several aspects. They frequently lack mechanisms for explainable intermediate reasoning, coherent agent collaboration, and alignment with intuitive human practices. Empirical research on human–agent interaction in innovation and strategic contexts, e.g., product development, innovation, and technology investment, remains scarce. Moreover, design principles for interfaces that support argumentation, iterative exploration, and mixed-initiative decision-making are not yet well established.

This dissertation addresses these gaps by developing a human-centered framework for leveraging LLMs and agentic AI in innovation and organizational decision-making. The research pursues four integrated aims. First, it empirically analyzes limitations in current decision and innovation practices through mixed-method studies involving managers, product development teams, and customer-facing innovation stakeholders. Second, this research examines how traditional multi-criteria analysis and opportunity evaluation methods can be integrated with LLM-assisted reasoning, synthetic user modeling, and agent-based workflows. Third, it designs and evaluates prototype architectures that enable collaboration between humans and AI agents in tasks such as idea generation, concept assessment, scenario exploration, and strategic prioritization. Fourth, it derives design principles, interaction

guidelines, and evaluation criteria to ensure explainability, transparency, and usability of such systems in organizational settings.

The dissertation contributes to management science, innovation research, and human-centered AI by demonstrating how LLMs and agentic AI can enhance analytical quality, creative exploration, and decision consistency across early and late-phase innovation processes. It also offers a structured foundation for developing transparent and trustworthy AI-assisted decision-support tools that augment, rather than replace, human judgment in complex organizational environments.